Network Policy

Peninsula Trauma Network

Peninsula Trauma Network Spinal Injury Standards

Date	Version
April 2015	V9

Purpose

Following the national introduction of Regional Trauma Networks, Major Trauma Centres (MTC's) are required to have a policy for the care of patients with traumatic spinal injuries.

The purpose of this policy is to provide direction and guidance for actions from key individuals and organisations within The Peninsula Trauma Network to improve the patient pathway and ensure that patients are transferred to the definitive point of care as quickly and safely as possible.

Who should read this document?

Trauma Network Clinical and Governance Directors
TU and MTC Clinical Leads for Major Trauma
Trauma Team Leaders
All Clinicians Transferring Patients to Specialist Centres
Acute Trust Lead Nurses

Key messages

Accountabilities

Production	Dr M Jadav
Review and approval	PTN Advisory Group
Ratification	PTN Executive Board
Dissemination	All PTN acute Trusts, South Western Ambulance Service NHS Foundation Trust, Devon Air Ambulance, Cornwall Air Ambulance, Search and Rescue Service.
Compliance	All Parties

Links to other policies and procedures

PTN Automatic Acceptance Policy
PTN Secondary Transfer Policy
PTN Safe Transfer of the Critically Ill Patient
Peninsula and Severn Trauma Networks Paediatric Secondary Transfer Policy

Version History

V8	Dr M Jadav	PTN Clinical Director
V9	Dr M Jadav	PTN Clinical Director

Last Approval	Due for Review
October 2015	October 2017

Peninsula Trauma Network Spinal Injury Standards

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Glossary of abbreviations

ASIA chart	American Spinal Injury Association International Standards For Neurological Classification Of Spinal Cord Injury
ED	Emergency Department
MTC	Major Trauma Centre
NDDH	North Devon District Hospital
PTN	Peninsula Trauma Network
R,D&E	Royal Devon & Exeter Hospital
RCH	Royal Cornwall Hospital
SCI	Spinal Cord Injury
SCIC	Spinal Cord Injury Centre
SSSC	Specialist Spinal Surgery Centre
SWASFT	Southwest Ambulance Service NHS Foundation Trust
TU	Major Trauma Unit

Aims

To ensure optimum care throughout the pathway from injury through rehabilitation for people suffering spinal cord or column injury within the Peninsula Trauma Network.

Background

Anecdotally, patients within the Peninsula Trauma Network have experienced unacceptable variability in the standard of care they receive following spinal injury, particularly in the areas of spinal clearance and referral.

In order to reduce variability and enable providers to develop a consistent approach to spinal injury care, this document has been produced by a regional multi-disciplinary, multi-agency working party. It is based upon the NSCISB 2011 guidance for trauma networks.

It is expected that the standards detailed within this document will form the basis for assessment of trauma care from providers within the Peninsula Trauma Network. As such, performance against these standards will be reportable to the Network and scrutinised during peer review assessments.

Assessment

Prehospital

Presumption of spinal injury

All people with a mechanism of injury consistent with spinal injury should be treated as having an unstable spinal injury until such injury is excluded.

All injured people complaining of pain in the spine, new neurological symptoms or exhibiting new neurological signs consistent with spinal cord injury should be treated as having an unstable spinal injury until such injury is excluded.

Spinal immobilisation

The SWASFT Spinal Immobilisation Policy is to be followed.

The patient with pre-existing spinal deformity or spinal disease should be immobilised in the position of greatest comfort rather than attempting to achieve the spinal neutral position.

Major trauma triage tool & primary transfer to major trauma centre or trauma unit

The Peninsula (Modified Wessex) Major Trauma Triage Tool is to be applied to all victims of trauma attended by prehospital emergency services.

The PTN Automatic Acceptance Policy as it pertains to spinal injury is to be followed: Primary transfer of patients with likely spinal injury and confirmed neurological deficit should be to the MTC, unless greater than 45 minute travel time or other factor requiring transfer to the nearest trauma unit or emergency department.

In ED

Documentation

Clinical examination of the whole spine and neurological status must be documented in all people with possible spinal injury.

Documentation of neurological assessment should be standardised by use of the ASIA chart in all people with a confirmed spinal cord injury.

Imaging

All spinal imaging should be reviewed and reported within 24hours of admission. [National Trauma Measure 2014]

Imaging of the cervical spine should be guided by one of the validated decision tools or national guidelines e.g. College of Emergency Medicine Clinical Effectiveness Committee Guideline on the management of alert, adult patients with potential cervical spine injury in the Emergency Department [http://secure.collemergencymed.ac.uk/code/document.asp?ID=5718]

Imaging of the thoracic or lumbar spine should be performed in all people who have:

A mechanism of injury consistent with spinal injury

Or

A predisposing condition placing them at high risk of pathological spinal injury e.g. ankylosing spondylitis, myeloma or bony metastases, osteoporosis or metabolic bone disorders.

And one of:

- Neurological symptoms or signs consistent with spinal cord or nerve root injury or compression
- Pain, Tenderness or new deformity in the thoracic or lumbar spine
- Inability to be adequately assessed due to reduced conscious level, or reduced appreciation of pain sensation (e.g. distracting injury, strong analgesia, intoxication, co-existing neuropathy).

When a significant spinal injury has been found, the whole of the spinal column should be imaged due to a high incidence of non-contiguous injury. [BOAST 2]

The choice of plain x-ray, CT and MRI should be as clinically indicated, if necessary in discussion with a spinal surgeon. Trauma (head, neck, thorax, abdomen, pelvis) CT should be requested as per the Peninsula Trauma Network Trauma CT protocol. In the presence of clinical signs of spinal cord injury or high energy mechanism of injury, CT of the affected area is the minimum imaging required. A finding of a spinal injury should be followed by the exclusion of non-contiguous injury. Whatever imaging is appropriate should be readily available within a time frame that does not compromise emergency management.

CT with senior radiologist report should be available 24 hours within 30min of arrival in the MTC and within 60min in the TUs.

MRI should be available at least weekdays in TUs and 24/7 in the MTC.

Where MRI is required more urgently than a TU can provide it, the patient will require emergency transfer to the MTC or specialised spinal surgery centre.

{For example, a patient with a possibly unstable injury demonstrated on CT and no neurological deficit may wait until the next day for further imaging. However, the patient with acute neurological deficit who requires time-critical MRI to plan surgical management may need to be transferred to the MTC or SSSC to receive this.}

Clearance

Clearance is defined as the exclusion of significant spinal column and / or spinal cord injury.

In patients of all ages, documented exclusion of significant spinal cord or column injury should precede mobilisation of the spine.

The cervical spine should remain in either "triple immobilisation" (collar, blocks and tape) or an appropriately-sized two-piece cervical immobilisation collar (Miami-j, Philadelphia, Aspen, etc.) until the cervical spine has been cleared.

Patients should remain in the most comfortable spinal-neutral position (supine or 40° lateral tilt with cervical alignment, never forcing neutral alignment where this increases pain or reverses chronic deformity) and log-rolled until thoracic and lumbar spine have been cleared. [BOAST 2]

Perform and document a clinical examination of the whole spine and neurological status (even if incomplete).

For all trauma patients who are anticipated to remain unconscious, unassessable or unreliable for more than 48 hours, radiological spinal clearance should be undertaken [BOAST 2]:

- 2-3mm slice helical CT scan of the cervical spine (base of skull to at least T1) with sagittal and coronal reconstructions. Extending this scan T4/5 overcomes the difficulties of imaging the upper thoracic spine.
- It is recommended that this cervical spine CT scan be undertaken as a routine with the first CT brain scan in all head-injured patients who have an altered level of consciousness.
- The remaining thoracic and lumbar spine may be adequately imaged either by AP and lateral plain radiographs or by sagittal and coronal reformatting of <5mm slice helical CT scans of the thorax, abdomen and pelvis undertaken as part of the trauma series.
- A senior radiologist must report spinal clearance images prior to removal withdrawal of spinal protection precautions.

Clinically assessable adults and older children (>9years old) may be cleared using a validated clinical decision rule (e.g. NEXUS / Canadian C-spine rule) and / or imaging.

There is insufficient evidence to recommend standards for spinal clearance in the youngest paediatric (0-3 year old) patients. Protocols including NEXUS or Canadian c-spine rules are difficult to apply, thus clinical judgement is used most frequently. [http://journals.lww.com/pccmjournal/Citation/2014/05001/ABSTRACT_786___CERVICAL_SPINE_CLEARANCE_IN_YOUNG.783.aspx]

Spinal column injury without spinal cord or cauda equina injury

Stability

All patients with spinal column injury without spinal cord injury should be assessed for referral:

1. "Insignificant" injury only: these are defined within the NEXUS and Canadian c-spine rules as per the table below. These do not require further discussion and can be managed conservatively. [Hoffman et al, N Engl J Med 2000; 343:94–99.]

"Insignificant" cervical spine injuries (Canadian C-spine rule):
• Isolated spinous process fracture not involving the lamina
• Simple vertebral compression fracture (<25% loss of height)
• Isolated osteophyte fracture (not corner or teardrop fracture)
• Isolated transverse process fracture not involving the facet joint

Patients with "insignificant" injuries should be assessed for range of motion and ability to bear physiological weight by sitting gradually upright. Severe pain indicates a more serious injury and requires continuation of spinal protection pending discussion with a spinal surgeon. Adequate range of motion and comfort in the upright position indicates initial stability. Patients with simple compression fractures and ongoing pain may still require follow-up weight-bearing imaging to detect progressive loss of vertebral height and all units should have local polices for this.

2. All other vertebral injuries: these should be discussed with a specialist spinal surgical centre (SSSC) or MTC to determine management.

General Management

Acute (minutes to hours)

Aims:

1. Prevention of progressive deformity
2. Prevention of progressive neurological deficit
3. Pain management
4. Prevention of non-spinal complications

Achieve these by:

1. The spine should not be stressed until anatomical and functional instability has been excluded.

2. As (1)

3.

3.1. Patients are encouraged to regularly report pain in a structured way
3.2. Staff and patients understand the relevance of pain to spinal instability and neurological injury
3.3. Staff record pain and provide optimal analgesia

4.

4.1. Pressure ulcer prevention
4.1.1. Patients should be regularly log-rolled to relieve pressure areas
4.1.2. Rigid cervical collars can also cause pressure ulceration: regularly provide manual in-line stabilisation of the cervical spine whilst releasing the collar and providing pressure area care.
4.2. Aspiration prevention
4.2.1. Patients unable to turn themselves safely require constant supervision by staff adequately equipped to clear the airway in case of vomiting.

Ongoing

Management as directed by a spinal surgeon.

Discharge

Discharge advice for patients with common injuries e.g. simple compression fractures of the thoracolumbar spine or cervical spine injury requiring hard collar should be available (example Appendix 2).

Spinal cord injury

Definition

Spinal Cord Injury (SCI) includes partial or full para or tetraplegia, and complete or incomplete cauda equina compression. This document refers only to traumatic SCI, however the spinal cord injury pathway is to be applied to patients with SCI of all aetiologies.

Presentation & Referral

Spinal Cord Injury presenting to an emergency department should be dealt with according to the algorithms in Appendix 3.

The Aims are:

1. To ensure reduction of low-velocity cervical SCI with dislocation within 4 hours of injury or presentation.
2. To ensure a specialist spinal surgeon (from R,D&E SSSC, Derriford MTC or Salisbury SCIC) agrees an optimum management plan within 4 hours of injury or presentation and this is documented within the medical notes.
3. To ensure that all patients are referred to the Salisbury SCIC within 24 hours of injury or presentation and their management plans agreed with the consultant at the SCIC.
4. To ensure that all patients have their details entered in the SCI registry database.
5. To ensure that all patients benefit from SCIC admission or outreach care.

Management

Acute (minutes to hours)

On identifying spinal cord injury, either clinically or radiologically, start the Spinal Cord Injury Pathway document (Appendix 4)

Aims:

1. Prevention of progressive deformity
2. Prevention of progressive neurological deficit
3. Pain management
4. Prevention of non-spinal complications

To achieve these objectives, the Spinal Cord Injury Pathway should be followed.

SCI Care Checklists(Appendix 5) should be available in all emergency departments to aid in the application of the optimal management plan.

Ongoing (hours to days)

Ongoing management of a patient with SCI should be closely guided by staff with expertise in spinal cord injury care.

There should be an optimised management plan recorded in the medical notes (i.e. the spinal cord injury pathway document) within 4 hours of injury. [NSCISB]

Protocols for nursing, joint protection and therapy requirements must be agreed with the SCIC.[BOAST 8]

Appropriately trained psychological support must be available for patient, family and carers.[BOAST 8]

If a patient is not transferred to the Salisbury SCIC, an outreach visit should be provided within 5 days of referral and then follow-up contact at least weekly [NSCISB].

Transfer

Peninsula Trauma Network Safe Transfer Policy is to be followed.

Rehabilitation

Patients with spinal column and/or cord injury will be assessed by appropriate therapy staff and a rehabilitation prescription will be commenced within 72hours of admission.

Audit and Monitoring

These standards should be audited by all Trauma Units and MTCs annually and the results presented to the Peninsula Trauma Network Executive Board prior to the annual Peer Review.

Appendix: 1 BOAST 2 & 8 standards

(https://www.boa.ac.uk/wp-content/uploads/2014/10/BOASTS-1-9.pdf)

BRITISH ORTHOPAEDIC ASSOCIATION

STANDARDS for TRAUMA (BOAST)

November 2008

BOAST 2: SPINAL CLEARANCE IN THE TRAUMA PATIENT

Background and Justification:
All patients involved in significant blunt trauma must be assumed to have an unstable injury to their spine; the incidence is approximately 2% and increases up to 34% in the unconscious patient. 50% of spinal injuries occur in the thoracic or lumbar spine; 20% at two levels. Immobilisation with full spinal precautions for prolonged periods creates difficulties in intensive care units. Spinal immobilisation is associated with pressure sores and pulmonary complications and is not recommended for more than 48 hours. Audits in the UK suggest poor implementation of spinal clearance policies. In the neck ligamentous disruption without a major bony injury may lead to instability. Recent comparative evaluations have shown that a modern helical CT scanning with reformatting can demonstrate the subtle abnormalities offering high sensitivity and specificity in detecting unstable injuries of the cervical spine. Plain radiographs are insensitive in the neck and the upper thoracic spine. MRI scanning has high sensitivity but only moderate specificity and is logistically difficult for ICU patients.

Inclusions: All trauma patients who are unconscious, unable to cooperate or who have distracting injuries that exclude reliable clinical assessment.

Exclusions: Children under the age of 16

Standards for Practice Audit:

1. A protocol for protection of the entire spine must be in place in all hospitals managing trauma patients at risk of spinal injury. This protection must be maintained from arrival until appropriate examination or investigations are completed and the spine cleared of injury.
2. Documentation of the neurological status must be made in all at-risk patients; any sign of spinal cord injury mandates urgent scanning.
3. A clinical examination of the whole spine should be documented.
4. If it is anticipated a patient will remain unconscious, unassessable or unreliable for clinical examination for more than 48 hours, radiological spinal clearance imaging should be undertaken.
5. For the cervical spine, the appropriate standard is a thin slice (2-3mm) helical CT scan from the base of the skull to at least T1 with both sagittal and coronal reconstructions; extending that scan to T4/5 overcomes the difficulties of imaging the upper thoracic spine.
6. It is recommended that this cervical spine CT scan be undertaken as a routine with the first CT brain scan in all head-injured patients who have an altered level of consciousness.
7. The remaining thoracic and lumbar spine may be adequately imaged either by AP and lateral plain radiographs or by sagittal and coronal reformatting of helical CT scans of the chest, abdomen and pelvis undertaken as part of a modern CT trauma series (<5mm slices).
8. A senior radiologist must report spinal clearance images prior to withdrawal of spinal protection precautions.
9. If a spinal injury is detected, a neurological assessment must be made, even if incomplete, and repeated regularly prior to urgent transfer to an appropriate spinal injury service.
10. MRI is the urgent investigation of choice for spinal cord injury.

Evidence Base:
Predominantly retrospective case series but with good expert reviews and an evolved multinational professional consensus over 15 years.

Limitations:
There are insufficient series or tested protocols to recommend a policy in children.
The place of MRI as a clearance tool for instability remains uncertain.
There are practical issues with scanning ICU patients and high false positive rates for intervertebral disc and ligament adnormality.

BASCIS BASS

BRITISH ORTHOPAEDIC ASSOCIATION

STANDARDS for TRAUMA (BOAST)

BOAST 8: THE MANAGEMENT OF TRAUMATIC SPINAL CORD INJURY

Background and Justification:

Spinal cord injury resulting in neurological deficit is a rare but potentially devastating injury. Compromise to the spinal cord may be due to trauma, vascular injury or other disease process and can result in immediate or insidious onset of neurological symptoms including loss or reduction of voluntary motor function, sensory impairment, bowel or bladder dysfunction and loss of autonomic function. The incidence in the United Kingdom is estimated at 12-16 per million population with about 75% of cases due to trauma. Appropriate management from the time of diagnosis of cord injury has been shown to have significant effect on the long-term outcome for patients and reduce short and long-term complications.

Included Patients:

All patients with traumatic spinal cord injury resulting in complete or incomplete para- or tetraplegia. The audit standards apply to those with polytrauma and those with isolated spinal cord injuries but do not apply to patients with spinal column injury without cord involvement. These audit standards apply to adults and children.

Standards for practice audit:

1. All Major Trauma Centres and Trauma Units must have a named, linked Spinal Cord Injury Centre.
2. All hospitals within a major trauma network should have an agreed, common protocol for protecting the neck and spine together with an agreed, common protocol to exclude significant injury (clearance of the neck and spine eg BOAST-2).
3. Centres managing patients with spinal cord injury require 24-hour access to CT and MRI.
4. Clinical evaluation of injured patients must include appropriate and repeated examination of the peripheral nervous system which should be recorded in the medical notes on an ASIA chart in keeping with the International Standards for Neurological Classification in Spinal Cord Injury.
5. Protocols for resuscitation and acute management including skin care, gastric, bowel and bladder care and neuroprotection must be agreed with the linked Spinal Cord Injury Centre and available in all Emergency Departments that may receive patients with spinal cord injury.
6. Centres treating these injuries must have the capability of performing specialist spinal surgery within 4 hours of injury. For those requiring surgery, protocols for anaesthesia and spinal stabilisation must be agreed with the linked Spinal Cord Injury Centre.
7. Protocols for nursing, joint protection and therapy requirements must be agreed with the linked Spinal Cord Injury Centre.
8. Management of the spine must follow written, agreed protocols with the linked Spinal Cord Injury Centre, or alternatively the on call consultant at that centre should be contacted within 4 hours of injury.
9. An early, joint management plan must be formulated and recorded in the medical notes within 12 hours.
10. Once the patient is fit for transfer to a Spinal Cord Injury Centre this should take place within 24 hours, unless it is agreed that it is the patient's best interest to remain in a Major Trauma Centre or Trauma Unit.
11. Spinal Cord Injury Centres should provide an outreach visit within 5 days of referral for patients unfit for transfer, and then follow-up contact (or visit) at least weekly until the patient is transferred.
12. Appropriately trained psychological support must be available for patients, family and carers.
13. All patients with new spinal cord injuries in England must have referral data submitted to the National Spinal Cord Injuries Database. The referral website is found at www.spinalcordinjury.nhs.uk

References:

References are found at www.spinalcordinjury.nhs.uk/docs.aspx

Management of People with Spinal Cord Injury. NHS Clinical Advisory Groups Report. (August 2011)

The Initial Management of Adults with Spinal Cord Injuries. (May 2012)

ASIA Protocol

Professional consensus

Appendix 2: Example of discharge advice sheets

For patients with thoracolumbar compression fractures

You have sustained a compression fracture of your _____ vertebra (back bone).

Normal bone is made up of a latticework of rigid fibres with spaces between them to allow blood cells and fluid to pass easily through. Your vertebra has lost some height by a compacting of this latticework. Your doctor has considered this to be a stable injury and you have been allowed to mobilise. It is important that you follow this advice to try to prevent further compaction of the bone which could lead to a deformity in your spine.

Your vertebra can carry your own weight but must not be placed under greater pressure than that until it has had time to heal.

Do try to stay mobile, using walking aids such as crutches or a frame if these have been given to you.

Do use regular pain killers. Your doctor suggests:

_____-

_____

Do not attempt to lift anything heavier than a newspaper until your back stops hurting.

Do not lift heavy objects for at least 6 weeks. When lifting, bend your knees and keep any load close to your body.

Do not bend your injured spine – bend at the knees to pick things up, or use a tool to reach.

Do not play impact sports such as tennis or squash or contact sports such as football or rugby until your doctor or physiotherapist advises you it is safe to do so (usually 8-12 weeks).

Do listen to your body – if you are getting increasing pain from an activity, stop.

You may have been given a follow-up appointment with a doctor or physiotherapist and you may require further x-rays periodically until the pain settles to check that the injured vertebra is not changing shape.

If you experience ongoing or worsening pain or swelling, or your ability to move worsens then seek advice from your doctor or physiotherapist.

If you have a further fall or injury and your pain worsens then attend the emergency department.

If you develop new weakness or numbness in your legs, or new problems with urination or opening your bowels, in particular a loss of sensation around the genitals or backside or a loss of control (incontinence) then see a doctor immediately or go to the emergency department.

For Patients with cervical injuries requiring a two-piece rigid cervical collar

You have sustained an injury to your neck bones (cervical spine).

The medical term for your injury is _____
_____.

In order to keep your neck in alignment and prevent any further damage to the spinal cord or nerves in the neck, you have been fitted with a two-piece rigid cervical collar.

(Delete as applicable:)

- You should wear your collar all the time apart from when performing the removal and cleaning procedure shown by your nurse or therapist..
- You may remove the collar at the following times _____
  _____.

The therapist or nurse who fitted the collar on you will have provided you with advice regarding the care of the collar and the skin beneath it. This information sheet tells you what to watch out for and who to contact if problems arise.

If you get any of the following problems, talk straight away to a doctor, either your own GP or attend the Emergency Department. In an emergency, call a 999 ambulance:

- Difficulty in breathing
- New numbness in the arms or legs
- Electric shocks going down your arms, back or legs.
- New weakness or clumsiness in the arms or legs.
- Increasing pain in the neck not settling with rest and simple pain-killers such as paracetmaol.

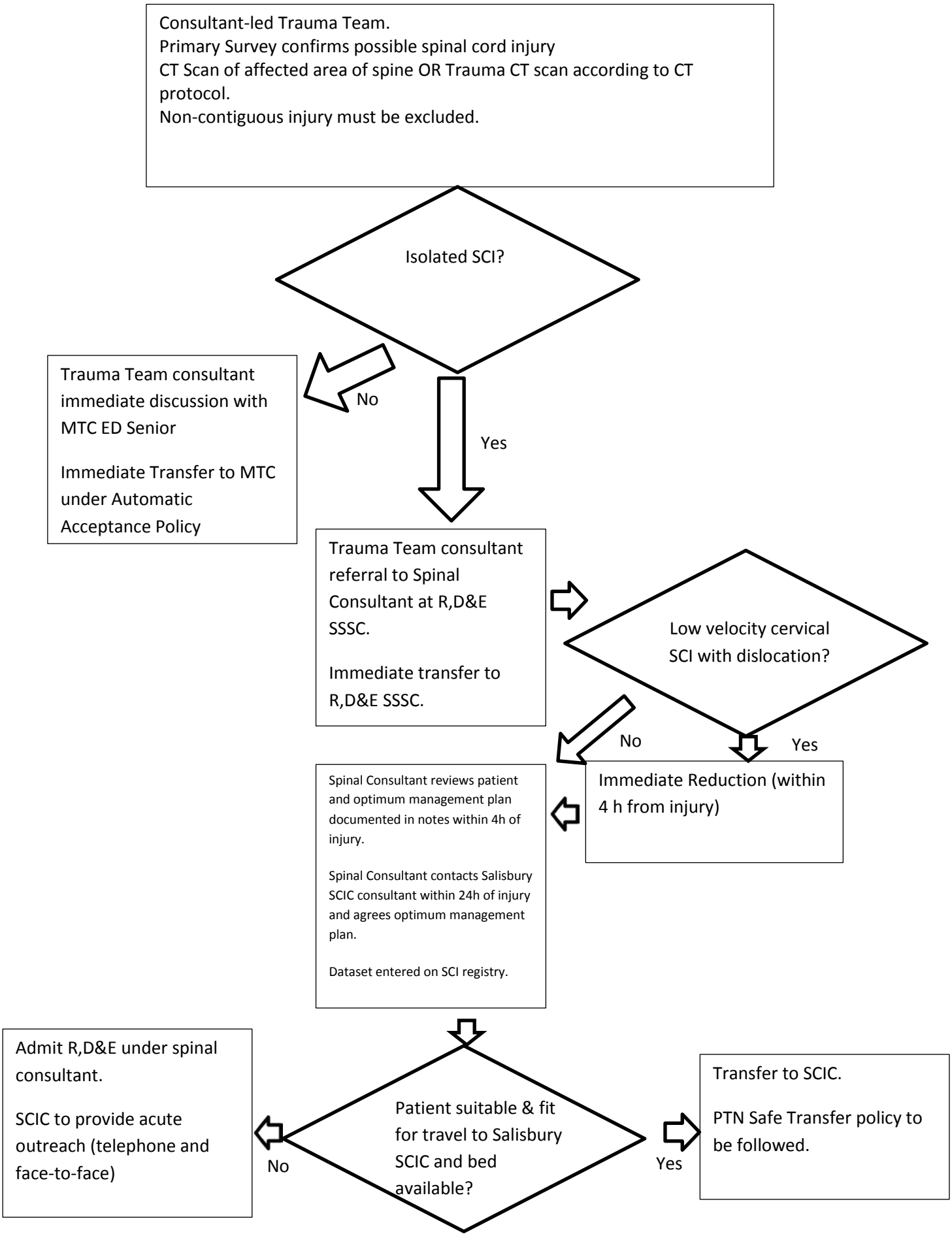
If you get any of the following problems, contact the person advised by the staff who fitted your collar, or your practice or district nurse:

- Soreness of the skin beneath the collar, or on the chin or chest.
- The collar fastenings are coming loose.
- The collar gets wet or dirty.
- You have something stuck under the collar.
- The collar does not seem to fit properly any more.

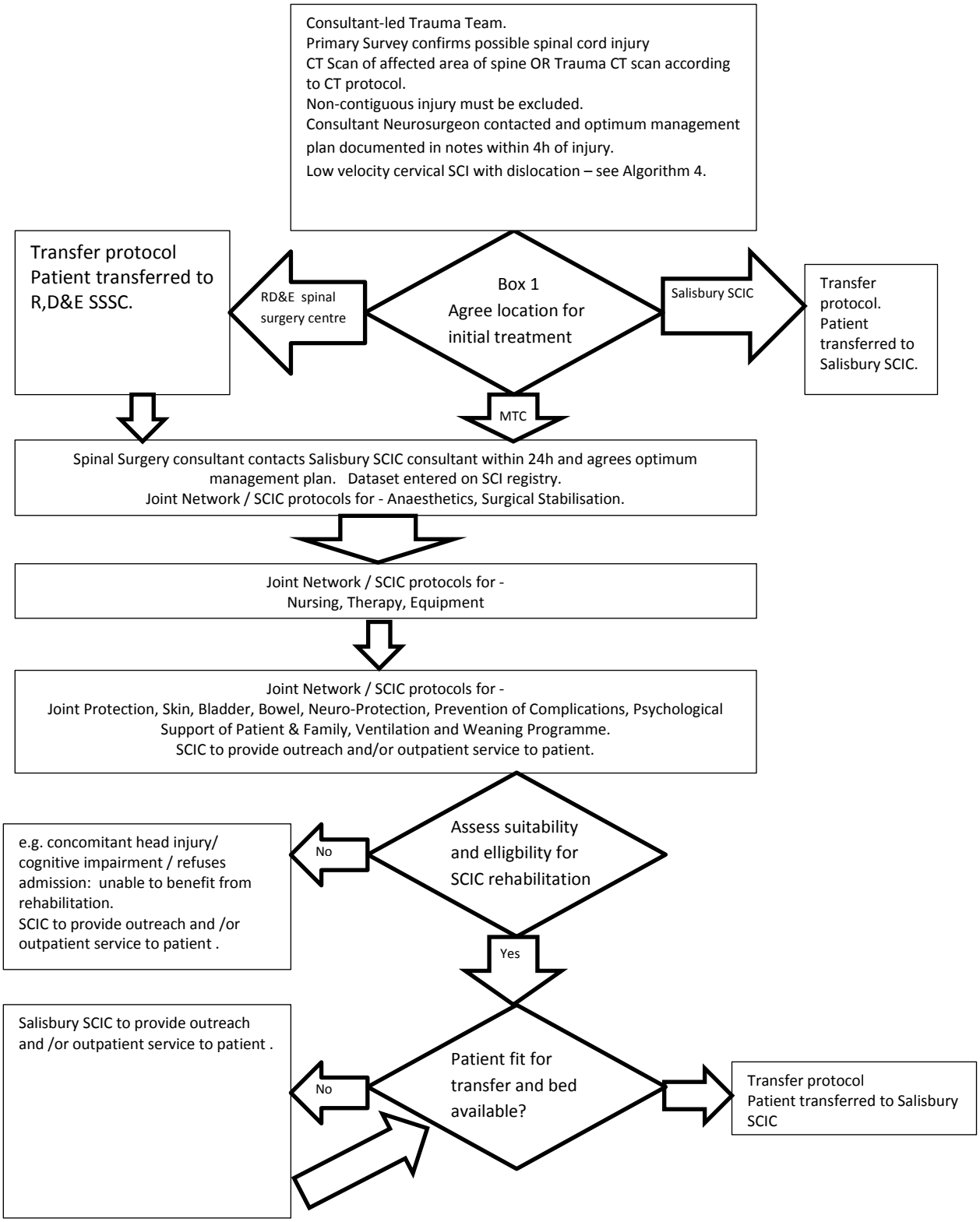
General concerns about your health or injury should be discussed with your GP.

Appendix 3 - Algorithms for acute presentation of Spinal Cord Injury to ED (adapted from NSCISB document)

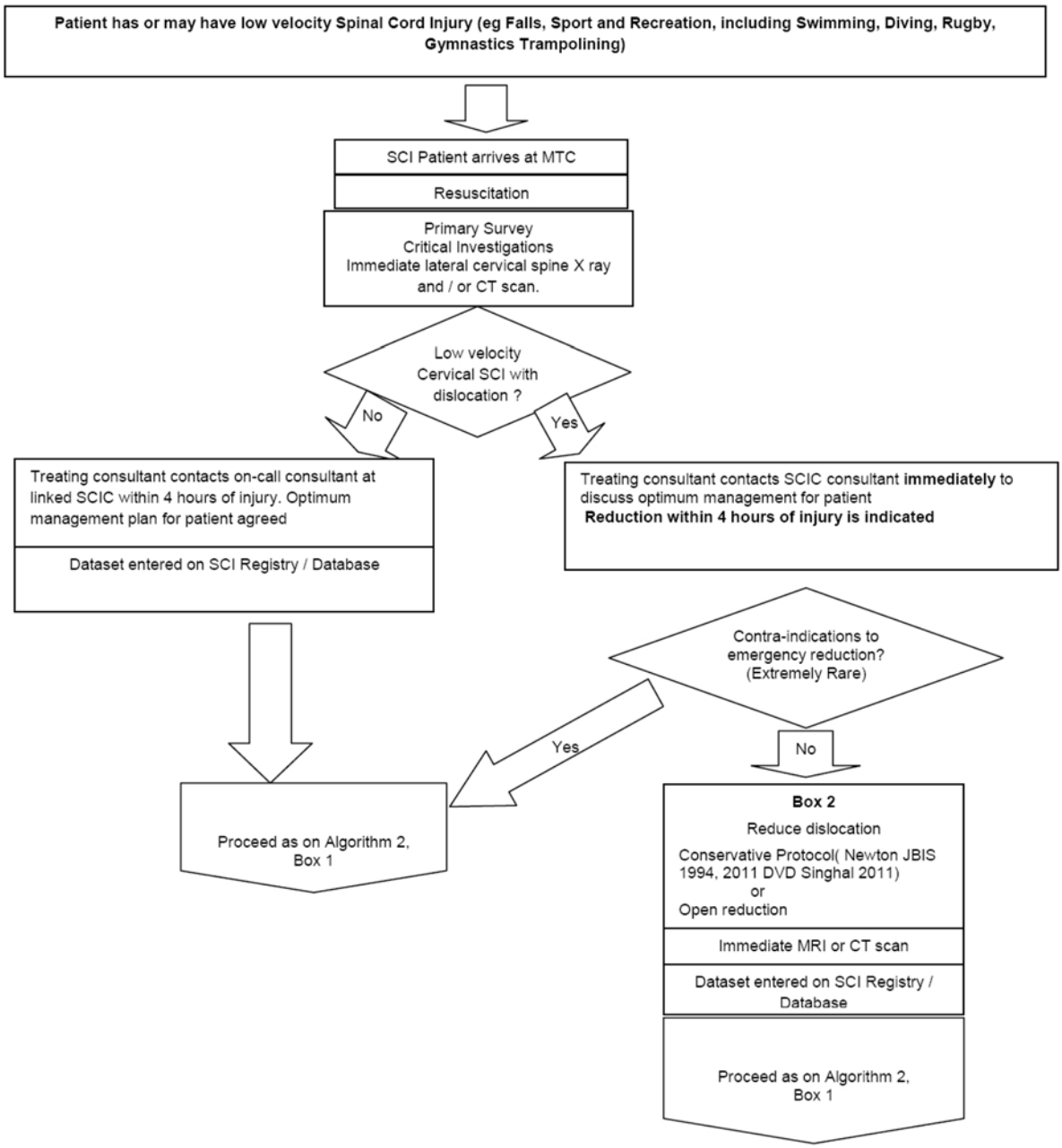
Algorithm 1 for Adult Patient attending NDDH Trauma Unit with Spinal Cord Injury



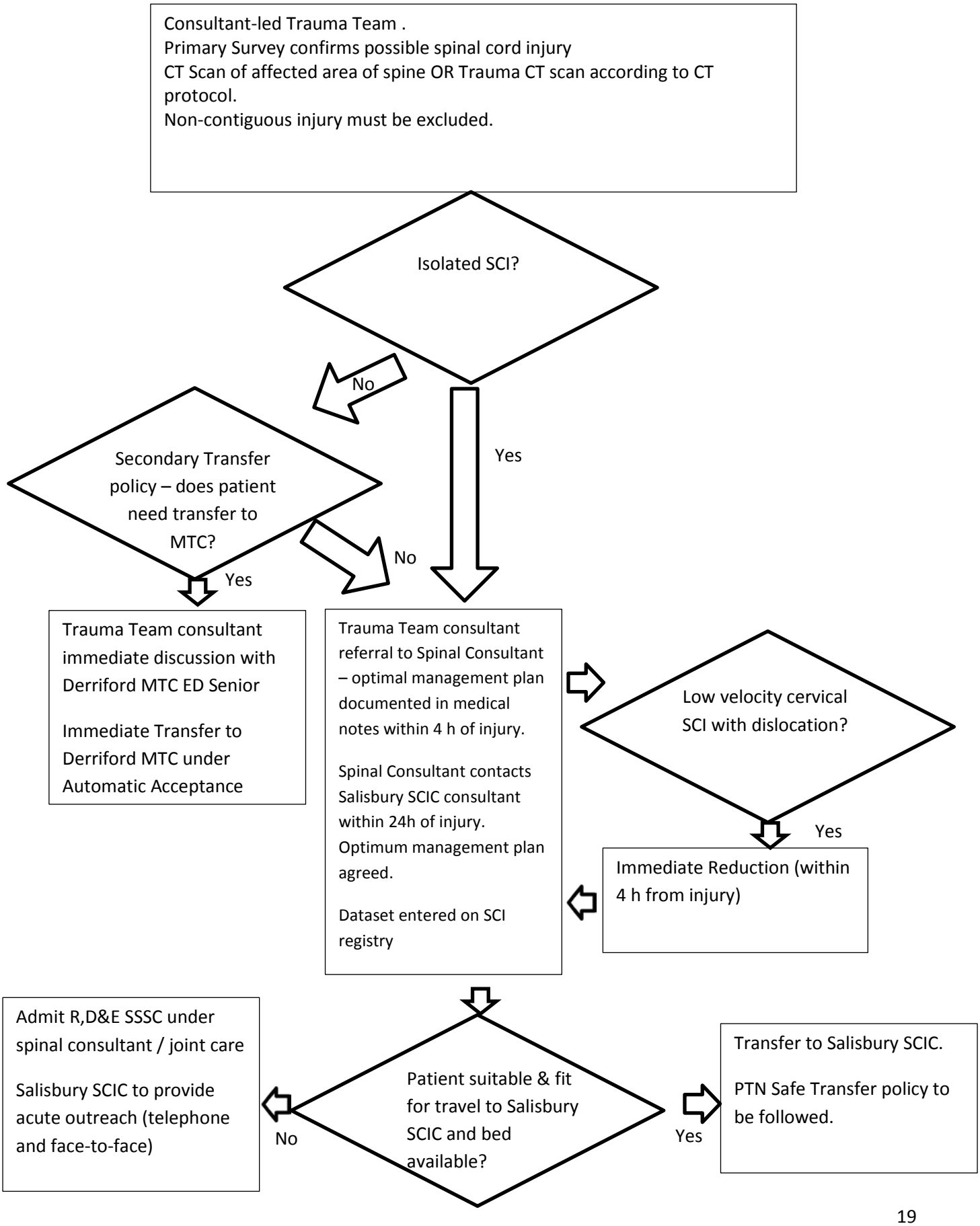
Algorithm 2 for Adult Patient with SCI attending Derriford Major Trauma Centre



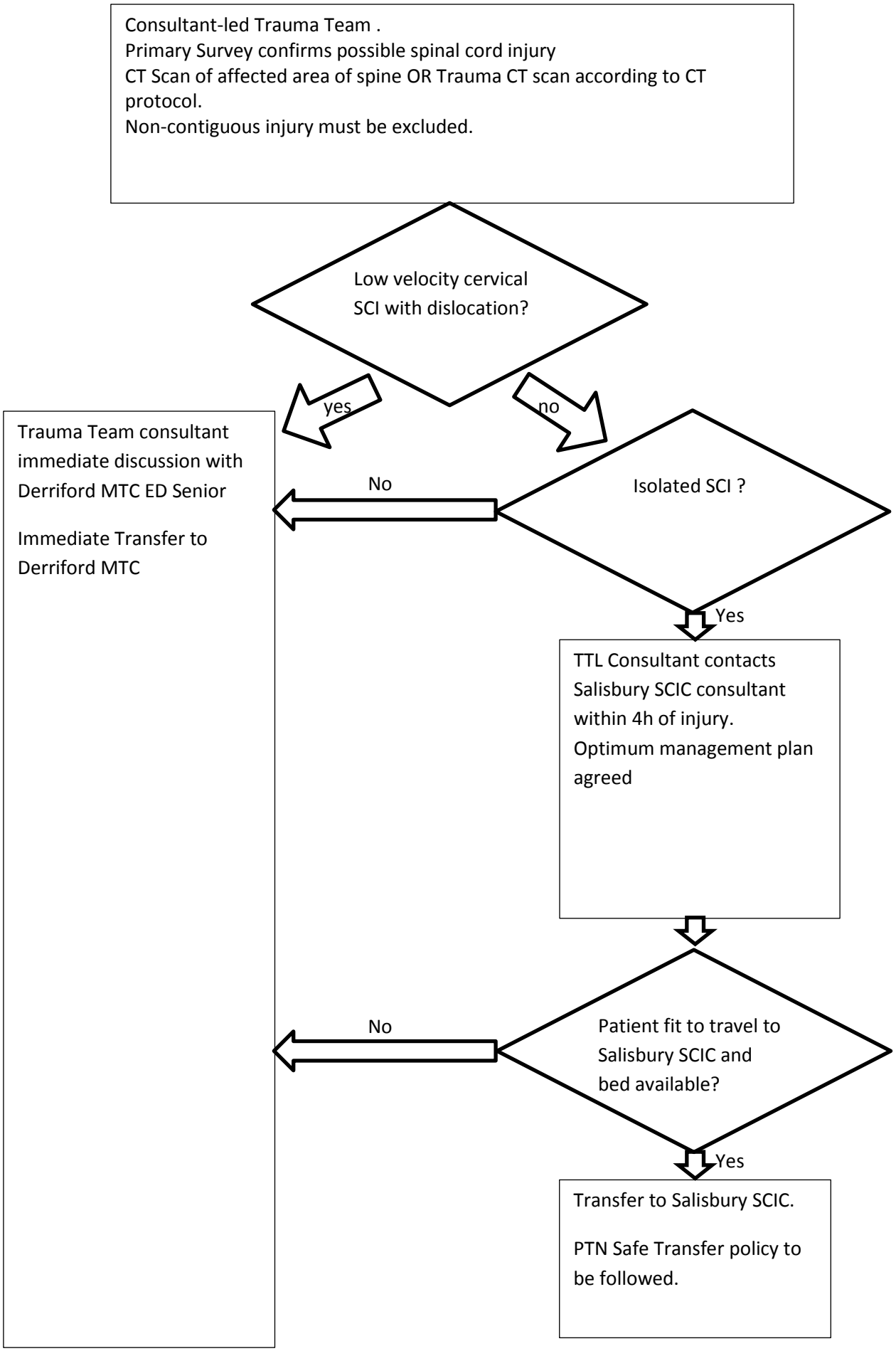
Algorithm 3 for Adult Patient attending Derriford MTC with Low Velocity Cervical SCI with Dislocation



Algorithm 4 for Adult Patient attending RD&E Specialist Spinal Surgery Centre & Trauma Unit with Spinal Cord Injury



Algorithm 5 for Adult Patient attending RCH or Torbay Trauma Units with Spinal Cord Injury



Appendix 4: Spinal Cord Injury Care Pathway

This document should be commenced as soon as spinal cord injury is diagnosed. The document is suitable for the management of spinal cord injury from all causes and has been developed by a regional multi-disciplinary group in conjunction with the Duke of Cornwall Spinal Treatment Centre.

Spinal Cord Injury Care Pathway

(In collaboration with Duke of Cornwall Spinal Treatment Centre, Salisbury District Hospital)

Patient name:..... DOB:.....Hospital Number:..... Address:..... ..... (or Addressograph)	**Consultant responsible for SCI care:** ..... *(Please complete the individual speciality clerking proforma for clinical details)*

Admission

Date of injury ..... Time of Injury.....

Mechanism of Injury: Traumatic SCI ☐ Non-traumatic SCI ☐

☐ Derriford Emergency Department Date ..... Time.....

☐ Transfer from..... ..Hospital Date..... Time .....

☐ Elective admission Date..... Time .....

Injury summary: Spinal & other injuries	Stable # / Unstable#	Initial Management Plan	Date

ASIA SCORE within 4 hours of admission			Neurological level of injury	Complete/ Incomplete	ASIA Impairment	Zone of partial preservation	
	Right	Left				Right	Left
Sensory							
Motor							
ASIA SCORE at 24 hours from admission			Neurological level of injury	Complete/ Incomplete	ASIA Impairment	Zone of partial preservation	
	Right	Left				Right	Left
Sensory							
Motor							
ASIA SCORE at 72 hours from admission			Neurological level of injury	Complete/ Incomplete	ASIA Impairment	Zone of partial preservation	
	Right	Left				Right	Left
Sensory							
Motor							
ASIA SCORE after surgery *(if applicable)*			Neurological level of injury	Complete/ Incomplete	ASIA Impairment	Zone of partial preservation	
	Right	Left				Right	Left
Sensory							
Motor							

1

Appendix 5: Spinal Cord Injury Care Plan

This document should be commenced at a specialist Spinal Treatment Centre. If a patient is transferred back to a local TU for ongoing care prior to transfer to the SCIC for specialist rehabilitation then this document should continue to be used by the therapists and nurses at the TU.

Multiprofessional care plan v12.pdf

Appendix 6: SCI Care Checklists

The following checklists have been drawn together by the MTC and are intended as an aide memoire for staff infrequently caring for patients with spinal cord injury, such as ED staff attending the acutely injured patient. The Please see the attached MTC Spinal Checklists document for further details.

The checklists are not a substitute for training and each ED or other area infrequently caring for patients with spinal cord injury should liaise with the Spinal Cord Injury Centre (Salisbury) to implement regular staff training and updates.

MTC Spinal Cord Injury Care Pathway V

Appendix 7: Contact details

Duke of Cornwall Spinal Treatment Centre, Salisbury District Hospital, Odstock, Salisbury. Tel: 01722 336262 (No paediatric service: Unable to admit <16 years)

Derriford Hospital MTC	0845 1155 8155 and page on-call Spinal Consultant
Royal Devon & Exeter Hospital SSSC	01392 411611 and page on-call Spinal Consultant

References

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NEXUS c-spine rule. N Engl J Med 2000; 343:94-99

Canadian C-Spine rule. N Engl J Med. 2003 Dec 25;349(26):2510-8.

Working Party

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